Integrity Level Mapping

Approximate correspondence between IEC 61508 SIL, ISO 26262 ASIL, and ISO 13849-1 PL · NOT equivalences

Integrity Level Correspondence

Approximate mapping — levels are derived differently and are NOT interchangeable

Integrity	IEC 61508	ISO 26262	ISO 13849-1	Risk Level	Target Measure
Highest	SIL 4	—	—	Catastrophic	PFH < 10^{9} /h
Very High	SIL 3	ASIL D	PL e	Life-threatening	PFH < 10^{8} /h
High	SIL 2	ASIL C	PL d	Severe injury	PFH < 10^{7} /h
Medium	SIL 1	ASIL B	PL c	Possible injury	PFH < 10^{6} /h
Low	—	ASIL A	PL b	Light injury	PFH < 10^{5} /h
Minimal	—	QM	PL a	Negligible	No target

CAUTION
These levels are derived by fundamentally different methods. SIL uses tolerable risk, ASIL uses S×E×C (HARA), and PL uses the risk graph (S×F×P). Direct mapping is approximate only.

Derivation Methods

How each standard determines its integrity level

IEC 61508

SIL 1–4

Tolerable Risk

- Tolerable risk target
- Consequence severity
- Frequency of exposure
- Possibility of avoidance
- Probability of occurrence

ISO 26262

QM, ASIL A–D

HARA (S × E × C)

- Severity (S0–S3)
- Exposure probability (E1–E4)
- Controllability (C1–C3)

ISO 13849-1

PL a–e (PLr)

Risk Graph (S × F × P)

- Severity (S1–S2)
- Frequency/duration (F1–F2)
- Possibility of avoidance (P1–P2)

Hardware Safety Metrics Comparison

Metric	IEC 61508	ISO 26262	ISO 13849-1
Dangerous failure rate	PFH (per hour)	PMHF (per hour)	PFH via MTTFd / DCavg / CCF
Diagnostic coverage	DC (SFF)	SPFM / LFM	DCavg (low / medium / high)
Architecture	HFT 0, 1, 2	ASIL decomposition	Category B, 1, 2, 3, 4
Common cause	Beta factor ()	ISO 26262-9 Annex D	CCF score (≥65 pts)

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